

Assignment

QR codes are provided for Video solutions of the questions. Please scan QR code to watch the solution.
 Questions to be discussed in Class.

SECTION - A

Objective Type Questions (One option is correct)

1. A particle moves on the curve $y = \frac{x^2}{4}$, where $x = t/2$. x and y are measured in metre and t in second. At $t = 4$ s, the velocity of particle is

- (1) $\frac{1}{\sqrt{2}}$ m/s
 (2) $\sqrt{3}$ m/s
 (3) $\sqrt{2}$ m/s
 (4) $2\sqrt{2}$ m/s



2. Position vector of a particle moving in a plane is given by $\vec{r} = (t^2 - 38t)\hat{i} + 2t^3\hat{j}$. The instant when velocity and acceleration of the particle are perpendicular to each other is

- (1) 1 s (2) 2 s
 (3) 3 s (4) 4 s



3. If the horizontal range of projectile be a and the maximum height attained by it is b , then velocity of projection is

- (1) $\left[g\left(b + \frac{a^2}{16b}\right)\right]^{\frac{1}{2}}$
 (2) $\left[2g\left(b + \frac{a^2}{16b}\right)\right]^{\frac{1}{2}}$
 (3) $\left[g\left(a + \frac{b^2}{16a}\right)\right]^{\frac{1}{2}}$
 (4) $\left[2g\left(a + \frac{b^2}{16a}\right)\right]^{\frac{1}{2}}$



4. The speed of a projectile when it is at its maximum height is $\sqrt{\frac{2}{5}}$ times its speed at half the maximum height. What is its angle of projection?

- (1) 30°
 (2) 45°
 (3) 60°
 (4) 90°



5. The horizontal range of a projectile is R and the maximum height attained by it is H . A strong wind now begins to blow in the direction of horizontal motion of projectile, giving it a constant horizontal acceleration equal to g . Under the same conditions of projection, the new range will be
 (g = acceleration due to gravity)

- (1) $R + H$ (2) $R + 2H$
 (3) $R + \frac{3H}{2}$ (4) $R + 4H$

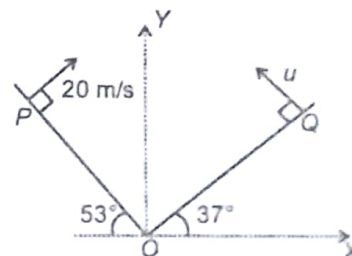


6. A ship is travelling due east at 10 km/h. Another ship heading 30° east of north is always due north from the first ship. The speed of second ship in km/h is

- (1) $20\sqrt{2}$ (2) $20\frac{\sqrt{3}}{2}$
 (3) 20 (4) $\frac{20}{\sqrt{2}}$

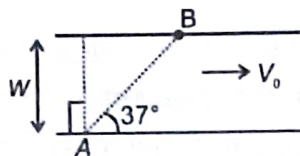


- 7*. Two particles are projected in vertical $X - Y$ plane from point P and Q as shown in figure. If they perform motion under gravity, then value of u so that collision take place during motion. $\left(\frac{OP}{OQ} = \frac{3}{4}\right)$



- (1) 20 m/s
 (2) $\frac{80}{3}$ m/s
 (3) 40 m/s
 (4) 15 m/s

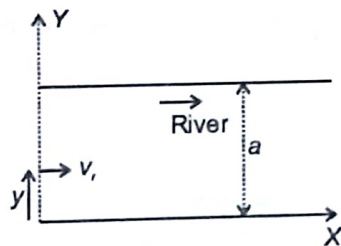
8. A swimmer A who want to reach directly at point B on the other bank of the river swims with minimum possible speed with respect to river and reaches directly at point B. If width of the river is w and river flows with velocity v_0 , then time in which swimmer crosses the river, is



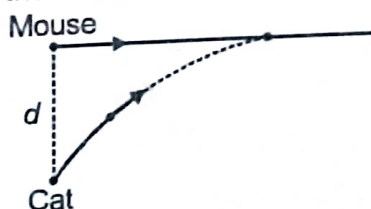
- (1) $\frac{25w}{16v_0}$ (2) $\frac{25w}{12v_0}$
(3) $\frac{16w}{25v_0}$ (4) $\frac{12w}{25v_0}$

9. A swimmer can swim at speed v_0 in still water. River flow speed is variable given by $v_r = \frac{v_0 y}{a}$, where y is the distance from one bank of the river and a is width of river. If swimmer crosses the river in shortest possible time then drift of the swimmer in this case is

- (1) a
(2) Zero
(3) $\frac{a}{2}$
(4) $\frac{a}{4}$



10. A mouse is at a distance of d from a cat. Both start running with constant equal speed v_0 in such a way that mouse moves in a straight line perpendicular to initial line joining cat and mouse, and cat always moves aiming at mouse. Separation between them when both move on the same line is

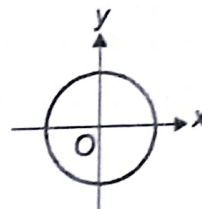


- (1) d (2) $2d$
(3) $\frac{d}{2}$ (4) Zero

11. A particle is performing uniform circular motion with speed π m/s. If radius of circular path is 2 m then magnitude of average velocity of particle in time interval, $\Delta t = 1.5$ s is

- (1) $\frac{2}{3}\sqrt{8+4\sqrt{2}}$ m/s (2) $\frac{1}{3}\sqrt{4+2\sqrt{2}}$ m/s
(3) Zero (4) $\frac{1}{3}\sqrt{2+2\sqrt{2}}$ m/s

12. An object moves at constant speed along a circular path in a horizontal xy plane, with center at origin. When the object is at $x = -2$ m, its velocity is $-4\hat{j}$ m/s. Its acceleration when it is at $y = 2$ m is



- (1) $-8\hat{i}$ m/s² (2) $-2\hat{j}$ m/s²
(3) $-8\hat{j}$ m/s² (4) $2\hat{i}$ m/s²

13. A particle is performing motion whose position vector is given by $\vec{r} = 10 (\cos pt \hat{i} + \sin pt \hat{j})$ m, when p is in rad/s and t is in seconds. The magnitude of centripetal acceleration at $t = 4$ s, is

- (1) $30p^2$ m/s²
(2) $10p^2$ m/s²
(3) $45p^2$ m/s²
(4) $90p^2$ m/s²

14. A particle is moving in X - Y plane on path $y = x^2 - 2x + 5$. Its x component of velocity is constant and is equal to 5 m/s. Acceleration of the particle at $x = 2$ m is

- (1) 25 m/s²
(2) 50 m/s²
(3) 10 m/s²
(4) 20 m/s²

15. Equation of trajectory of a projectile is given as $y = x - \frac{x^2}{40}$. The radius of curvature at highest point of its trajectory is [Take, $g = 10$ m/s²]

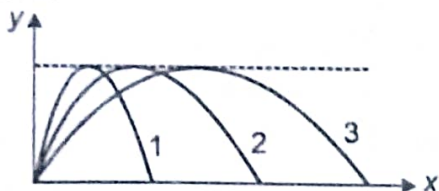
- (1) 10 m
(2) 20 m
(3) 40 m
(4) 5 m

SECTION - B

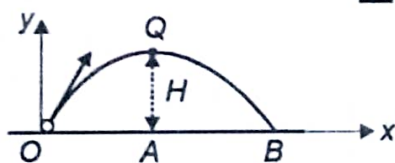
Objective Type Questions

(One or more than one option(s) is/are correct)

1. The figure shows three paths for a ball projected from ground level. Ignore the effects of air on flight. Which of the following statements is correct?



- (1) The projectile 3 has greatest time of flight
 (2) The initial vertical velocities are same for all
 (3) The projectile 3 has greatest speed of projection
 (4) The initial horizontal velocities are same for all
2. Figure shows the path of a projectile thrown near the surface of earth in the absence of air resistance.



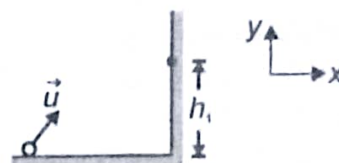
If the air resistance is taken into consideration, then

- (1) The time taken to move from O to Q is greater than the time taken to move from Q to B
 (2) At Q, acceleration of projectile is greater than g
 (3) H decreases
 (4) Speed of the projectile at B is equal to the speed of projection
3. Two particles are projected from same point with same initial speed u at angles of projection α and β . t_1 and t_2 are their respective time of flights and h_1 and h_2 are their respective maximum heights. If they strike the horizontal ground at same point having range R , then



- (1) $R = 4\sqrt{h_1 h_2}$ (2) $\alpha + \beta = \frac{\pi}{2}$
 (3) $\tan \alpha = \frac{t_1}{t_2}$ (4) $\tan \alpha = \sqrt{\frac{h_1}{h_2}}$

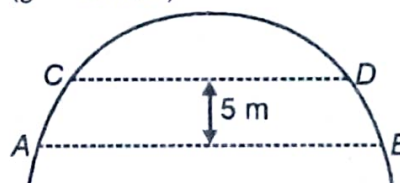
4. A ball is thrown with velocity $\vec{u} = (3\hat{i} + 4\hat{j})$ m/s towards a wall, where it hits at height h_1 at time t_1 after the launch.



Another ball is projected from same point with velocity $\vec{u} = (3\hat{i} + 5\hat{j})$ m/s towards the wall where it hits at height h_2 at time t_2 after the launch. Then



- (1) $t_1 > t_2$
 (2) $t_1 = t_2$
 (3) $h_2 > h_1$
 (4) No assertion can be made for h_1 and h_2
5. A particle is projected from ground to ground and its path is as shown in the figure. Points A and B are at same horizontal level and points C and D are at another same horizontal level. Separation between line AB and CD is 5 m. If particle takes time t_1 to go from A to B and time t_2 from C to D, then is ($g = 10 \text{ m/s}^2$)

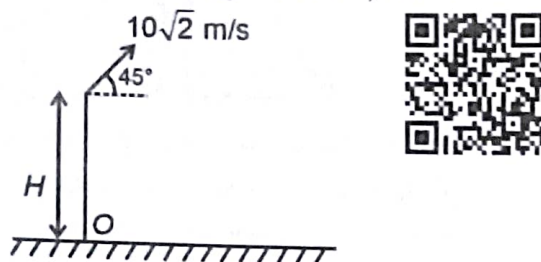


- (1) $(t_1^2 - t_2^2) = 8 \text{ s}^2$
 (2) $(t_1^2 - t_2^2) = 4 \text{ s}^2$
 (3) $(t_1 + t_2) = 8 \text{ s}$
 (4) $(t_1 - t_2) = 4 \text{ s}$
6. The equation of trajectory of a projectile thrown from a level ground near the surface of earth is given by $y = ax - bx^2$, with y-axis in vertical direction and x-axis in horizontal direction. a and b are constants. Then
- (1) The range of the projectile is $\frac{a}{b}$
 (2) At $x = \frac{a}{2b}$, the velocity of projectile becomes zero
 (3) The maximum height attained by projectile is $\frac{a^2}{4b}$
 (4) The angle of projection is $\tan^{-1}(a)$

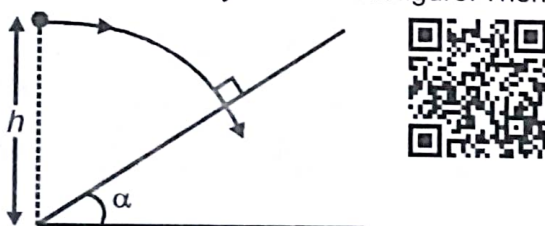


7. A particle is projected with speed $10\sqrt{2}$ m/s from top of a tower of height H as shown in figure

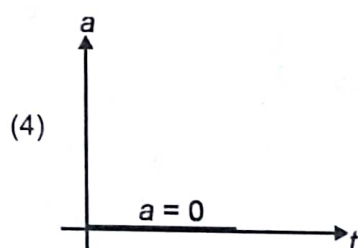
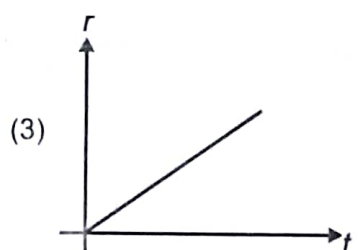
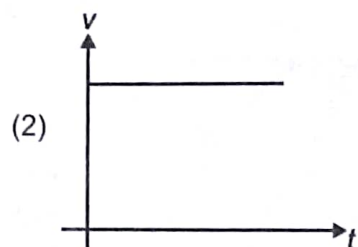
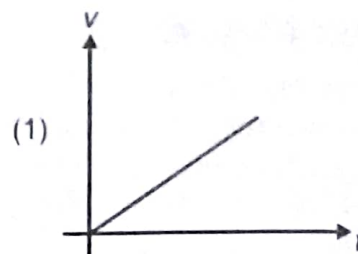
(Take $H = 10$ m and $g = 10$ m/s²). Then



- (1) After time $t = 2$ s its velocity becomes perpendicular to its initial velocity
 - (2) Speed of the particle just before collision with the horizontal surface will be 20 m/s
 - (3) Time of flight will be $(\sqrt{3} - 1)$ s
 - (4) Particle will collide with the horizontal surface at an angle of 60° with the horizontal
- 8*. A stone is projected horizontally with velocity v from a building of height h . It hits the inclined plane perpendicularly as shown in figure. Then



- (1) Time taken to hit the plane is $\frac{v}{g \tan \alpha}$
 - (2) Time taken to hit the plane is $\frac{v \tan \alpha}{g}$
 - (3) The value of v is $\sqrt{\frac{gh}{2 - \cot^2 \alpha}}$
 - (4) The value of v is $\sqrt{\frac{2gh}{2 + \cot^2 \alpha}}$
9. Two projectiles are projected from same position under gravity at different angles with horizontal in the same vertical plane. If velocity of one with respect to other is v , acceleration of one with respect to other is a and position of one with respect to other is r , then which of the following graphs is(are) correct?



10. Two particles are projected from the same point, with the same speed, in the vertical plane, at different angles with the horizontal. A frame of reference is fixed to one particle. The position vector of the other particle, as observed from this frame is $\vec{r}$. Then



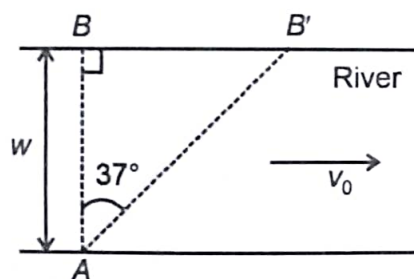
- (1) The direction of $\vec{r}$ changes with time
- (2) The magnitude of $\vec{r}$ remains constant with time
- (3) The direction of $\vec{r}$ remains same with time
- (4) The magnitude of $\vec{r}$ increases linearly with time

11. A man can swim with a velocity v relative to water. He has to cross a river of width d flowing with a velocity u ($u < v$). The distance through which he is carried downstream by the river is x . Choose the correct statement.



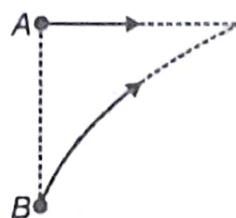
- (1) For x to be minimum, he has to swim in a direction making an angle $(\frac{\pi}{2} + \sin^{-1} \frac{u}{v})$ with the flow
- (2) For x to be maximum, he has to swim in a direction making an angle of $(\frac{\pi}{2} - \sin^{-1} \frac{u}{v})$ with the flow
- (3) If $x = 0$, the time taken to cross the river is $\frac{d}{\sqrt{v^2 - u^2}}$
- (4) $x = \frac{du}{v}$ if he takes minimum time to cross the river

- 12*. A swimmer at a point A on the river bank wants to cross it. River is flowing with speed v_0 . (Refer the diagram below). If v_{sr} denotes velocity of swimmer in still water, then



- (1) In order to reach directly at point B, v_{sr} must be greater than v_0
- (2) In order to reach directly at point B', $v_{sr} \geq (\frac{4v_0}{5})$
- (3) If $v_{sr} = 2v_0$, then swimmer must head upstream at an angle, $\theta = 30^\circ$ with the line AB to reach directly at point B
- (4) If swimmer reaches directly at point B' with minimum possible v_{sr} then time taken by swimmer to cross the river of width w will be $(\frac{25w}{12v_0})$

13. Two particles A and B are at rest at separation of 160 m. Both the particle starts moving at $t = 0$. Particle A moves in a straight line perpendicular to initial position AB with constant acceleration of 3 m/s^2 . Particle B moves in such a way that it always aims towards A and moves with speed increasing at constant rate of 5 m/s^2 . Then



- (1) The time elapsed when both particles meet is 5 s
- (2) The time elapsed when both particles meet is 10 s
- (3) Distance covered by A before meeting is 150 m
- (4) Distance covered by A before meeting is 200 m

14. Which of the following statements is/are correct?



- (1) The net acceleration of a particle in circular motion is always towards the centre of the circle
- (2) The velocity vector of a particle at a point is always along the tangent to the path of the particle at that point
- (3) The acceleration vector of a particle in uniform circular motion averaged over one cycle is zero
- (4) It is not possible to revolve a particle in circular path with constant acceleration

15. A particle starts performing circular motion in a circle of radius 1 m at $t = 0$. Its speed v varies with time t as, $v = 4t \text{ m/s}$, where t is in seconds. If a_r and a_t are the radial and tangential acceleration of the particle respectively, then

- (1) $a_t = 1 \text{ m/s}^2$
- (2) $a_t = 4 \text{ m/s}^2$
- (3) $a_r = 16 \text{ m/s}^2$ at $t = 1 \text{ s}$



- (4) At $t = \frac{1}{2} \text{ s}$, its velocity vector makes an angle of 45° with acceleration vector

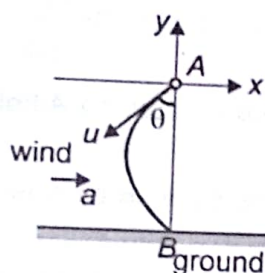
SECTION - C

Linked Comprehension Type Questions

This section contains 3 paragraphs. Based upon each paragraph, some multiple choice questions have to be answered. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Comprehension-I

A particle is thrown with speed u from A above the ground at an angle θ with the vertical as shown in the figure. A strong horizontal wind gives the particle a constant acceleration a in the $+x$ direction. If the particle strikes the ground at a point B directly below A.

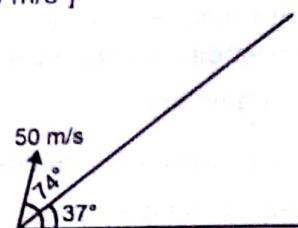


Choose the correct answer :

- The time taken by the particle to reach to B is
 (1) $\frac{2u \sin \theta}{g}$ (2) $\frac{2u \cos \theta}{g}$
 (3) $\frac{2u \sin \theta}{a}$ (4) $\frac{2u \cos \theta}{a}$
- The maximum distance of the particle from the line AB is
 (1) $\frac{u^2 \sin^2 \theta}{2a}$ (2) $\frac{u^2 \sin^2 \theta}{2g}$
 (3) $\frac{u^2}{2\sqrt{g^2 + a^2}}$ (4) $\frac{u^2 \sin^2 \theta}{2\sqrt{g^2 + a^2}}$
- The distance AB is
 (1) $\frac{u^2 \sin 2\theta}{a}$
 (2) $\frac{u^2 \sin 2\theta}{g}$
 (3) $\frac{2u^2}{g} \sin \theta \left(\cos \theta + \frac{a}{g} \sin \theta \right)$
 (4) $\frac{2u^2}{a} \sin \theta \left(\cos \theta + \frac{g}{a} \sin \theta \right)$

Comprehension-II

A particle is projected up an inclined plane at angle 74° with horizontal with speed 50 m/s under gravity. [Take, $g = 10 \text{ m/s}^2$]

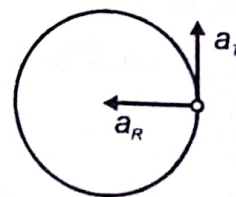


Choose the correct answer :

- Time taken by the particle to strike the inclined plane is
 (1) 7.5 s (2) 15 s
 (3) 10 s (4) 5 s
- Range along the inclined plane is
 (1) $\frac{125}{4} \text{ m}$ (2) $\frac{375}{4} \text{ m}$
 (3) $\frac{525}{4} \text{ m}$ (4) $\frac{625}{4} \text{ m}$
- The ratio of range along the inclined plane and maximum height above the inclined plane is
 (1) $\frac{7}{3}$ (2) $\frac{7}{2}$
 (3) 7 (4) $\frac{7}{4}$

Comprehension-III

A particle moves with decreasing speed along the circle of radius R so that at any moment of time its tangential and centripetal accelerations are equal in magnitude. At the initial moment, $t = 0$, its speed is u .



Choose the correct answer :

- The time after which the speed of particle reduces to half of its initial value is
 (1) $\frac{2R}{u}$ (2) $\frac{R}{u}$
 (3) $\frac{R}{2u}$ (4) $\frac{3R}{2u}$

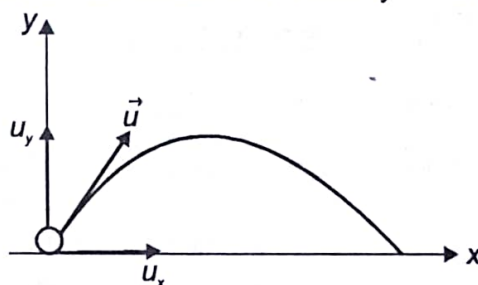
2. The magnitude of tangential acceleration at $t = \frac{R}{2u}$ is
- (1) $\frac{u^2}{R}$ (2) $\frac{2u^2}{3R}$
 (3) $\frac{4u^2}{9R}$ (4) $\frac{7u^2}{36R}$
3. The distance travelled by the particle when its speed becomes 37% of its initial value is [Take $e^{-1} = 0.37$]
- (1) R (2) $2R$
 (3) $2\pi R$ (4) πR

SECTION - D

Matrix-Match Type Questions

This section contains 3 questions. Each question contains statements given in two columns which have to be matched. The statements in Column-I are labelled A, B, C and D, while the statements in Column-II are labelled p, q, r, s and t. Any given statement in Column-I can have correct matching with One OR More statement(s) in Column-II.

1. A particle is projected from ground with initial velocity $\vec{u} = u_x\hat{i} + u_y\hat{j}$ as shown in the figure. If a_x and a_y are the components of acceleration in horizontal and vertically downward directions then



Match the following columns.

Column-I

- (A) Time of flight depends on
 (B) Maximum height depends on
 (C) Range depends on
 (D) Speed at the top depends on

Column-II

- (p) u_x
 (q) u_y
 (r) a_x
 (s) a_y

2. Column-I contains the description of a particle in projectile motion under gravity or in circular motion while Column-II contains the properties of respective motion, then match the column-I with column-II.

Column-I

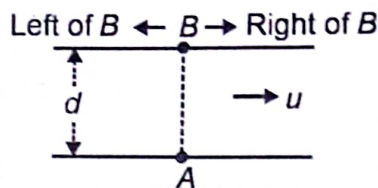
- (A) Remains constant in projectile motion
 (B) Does not remain constant in uniform circular motion
 (C) Remains constant in uniform circular motion
 (D) Does not remain constant in projectile motion

Column-II

- (p) Centripetal acceleration
 (q) Tangential acceleration
 (r) Speed
 (s) Velocity
 (t) Acceleration



3. A river of width d is flowing with speed u as shown in the figure. John can swim with maximum speed v relative to the river and can cross it in shortest time T . John starts at A . B is the point directly opposite to A on the other bank of the river. If t be the time John takes to reach the opposite bank, match the situation in the column-I to the possibilities in column-II.



Column-I

- (A) John reaches to the left of B
 (B) John reaches to the right of B
 (C) John reaches the point B
 (D) John drifts along the bank while minimizing the time

Column-II

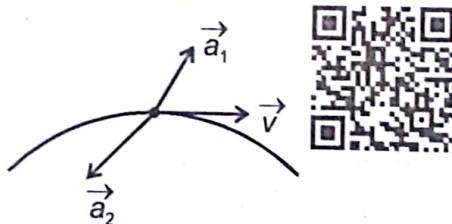
- (p) $t = T$
 (q) $t > T$
 (r) $u < v$
 (s) $u > v$

SECTION - E

Assertion-Reason Type Questions

This section contains 5 questions. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (2) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
 (3) Statement-1 is True, Statement-2 is False
 (4) Statement-1 is False, Statement-2 is True
1. A body moves in a curved path as shown.



STATEMENT-1 : Its acceleration vector cannot be $\vec{a}_1$, but it may be $\vec{a}_2$.

and

STATEMENT-2 : The normal acceleration is always directed towards the centre of curvature.

2. STATEMENT-1 : A particle is projected with speed u at an angle θ with the horizontal. At any time during motion, speed of particle is v at angle α with the vertical, then $v \sin \alpha$ is always constant throughout motion.

and

STATEMENT-2 : In projectile motion, magnitude of radial acceleration at topmost point is minimum.

3. STATEMENT-1 : A food packet is dropped from a rescue plane moving with constant velocity. Path of the food packet will be straight line for the pilot but parabolic for the person on the ground if air resistance is negligible.

and

STATEMENT-2 : Food packet has initial velocity same as that of plane.

4. STATEMENT-1 : A man standing on the ground has to hold his umbrella at an angle θ with the vertical to protect himself from rain. It is possible that the man now starts running on the ground with certain speed, but he still has to hold his umbrella at angle θ with the vertical to protect himself from rain.



and

STATEMENT-2 : Vertical component of velocity of rain w.r.t. man does not change, if he starts running on a horizontal ground.

5. STATEMENT-1 : In circular motion, acceleration may be in any direction relative to the velocity.



and

STATEMENT-2 : Dot product of velocity and acceleration in circular motion may be positive, negative or zero.

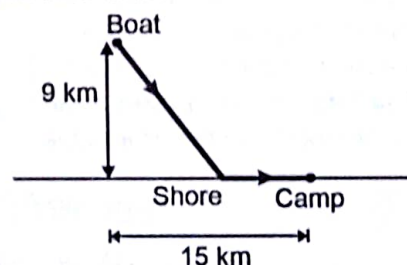
3. A particle is projected horizontally from top of a tower. If v is the velocity at any instant and radius of curvature of the path of the particle at that instant is directly proportional to v^n , then find the value of n .



4. A fishing boat is anchored 9 km away from the nearest point on the shore. A messenger must be sent from the fishing boat to a camp, 15 km from the point on shore closest to boat.



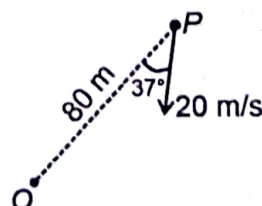
If the messenger can walk at a speed of 5 km per hour and can row at 4 km/h, determine the distance of that point (in km) from the camp, where he must land so as to reach the camp in least possible time.



5. A man walking with a speed of 3 km/h finds the rain drops falling vertically downwards. When the man increases his speed to 6 km/h he finds that the rain drops are falling making an angle of 30° with the vertical. Find the speed of the rain drops (in km/h).



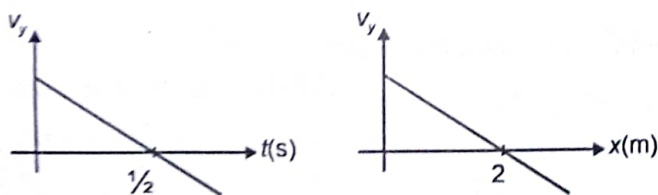
- 6*. P is a movable point moving with constant speed 20 m/s such that its velocity vector always maintains an angle 37° with line OP . If initial distance between O and P is 80 m. After what time (in s) shall P reach O ?



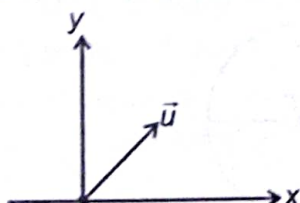
SECTION - F

Integer Answer Type Questions

- 1*. The two graphs of a given projectile show that variation of vertical velocity component with time and with horizontal displacement. What is the x -component of velocity in m/s?

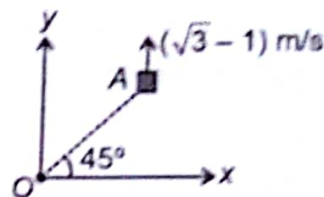


2. A projectile is given an initial velocity of $\vec{u} = (\hat{i} + 2\hat{j})$ m/s in vertical plane as shown. If equation of its trajectory is $y = ax - bx^2$ then value of $(a + b)$ is equal to ($g = 10 \text{ m/s}^2$)

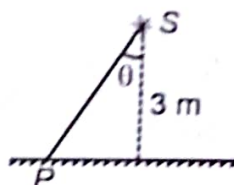


7. Four persons K, L, M, N are initially at the four corners of a square of side 20 m. Each person now moves with a uniform speed $v = 5$ m/s in such a way that K always moves directly towards L , L directly towards M , M directly towards N and N directly towards K . Find time (in second) after which four persons will meet.
8. Six particles are situated at the corners of a regular hexagon. These particles start moving with equal speed of 10 m/s in such a way that velocity of any one particle is directed towards the next particle. Find the rate (in m/s) at which length of a side of the hexagon is decreasing.
9. On a frictionless horizontal surface (x - y plane) a small trolley A is moving along a straight line parallel to y -axis with constant speed of $(\sqrt{3} - 1)$ m/s. A ball is thrown along the surface from the origin O when trolley was

at A . Its velocity makes an angle ϕ with the x -axis and it hits the trolley. If $\phi = \frac{4\theta}{3}$, where θ is the angle made by the velocity vector of the ball with x -axis in the frame of trolley, find the speed (in m/s) of the ball with respect to the surface.



10. Spotlight S rotates in a horizontal plane with constant angular velocity of 0.1 rad/s. The spot of light P moves along the wall at a distance of 3 m. If the velocity of spot P when $\theta = 45^\circ$ is v then value of $10v$ in m/s is equal to



SECTION - G (Previous Years Questions)

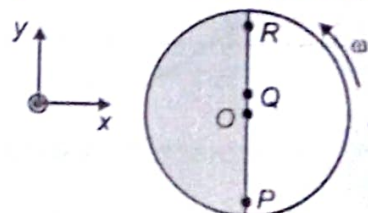
Only One Option Correct Type Questions

- 1*. Consider a disc rotating in the horizontal plane with a constant angular speed ω about its centre O . The disc has a shaded region on one side of the diameter and an unshaded region on the other side as shown in the figure. When the disc is in the orientation as shown, two pebbles P and Q are simultaneously projected at an angle towards R . The velocity of projection is in the y - z plane and is same for both pebbles with respect to the disc. Assume that (i) they land back on the disc before the disc has completed $\frac{1}{8}$ rotation, (ii) their range is less than half the disc radius, and (iii) ω remains constant throughout. Then



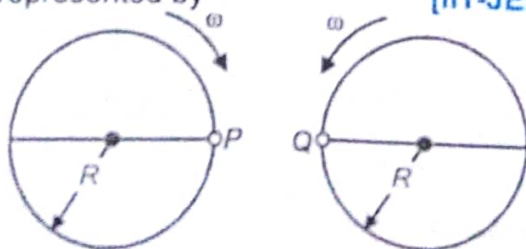
- (A) P lands in the shaded region and Q in the unshaded region
 (B) P lands in the unshaded region and Q in the shaded region
 (C) Both P and Q land in the unshaded region
 (D) Both P and Q land in the shaded region

[IIT-JEE-2012 (Paper-2)]



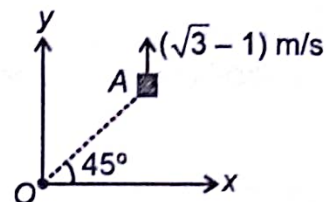
- 2*. Two identical discs of same radius R are rotating about their axes in opposite directions with the same constant angular speed ω . The discs are in the same horizontal plane. At time $t = 0$, the points P and Q are facing each other as shown in the figure. The relative speed between the two points P and Q is v_r . In one time period (T) of rotation of the discs, v_r as a function of time is best represented by

[IIT-JEE-2012 (Paper-2)]

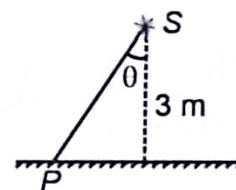


7. Four persons K, L, M, N are initially at the four corners of a square of side 20 m. Each person now moves with a uniform speed $v = 5$ m/s in such a way that K always moves directly towards L , L directly towards M , M directly towards N and N directly towards K . Find time (in second) after which four persons will meet.
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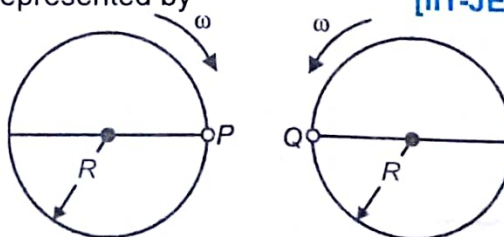
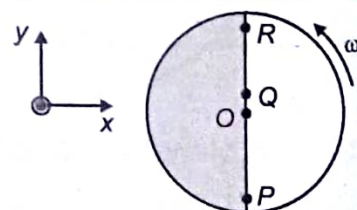


SECTION - G (Previous Years Questions)

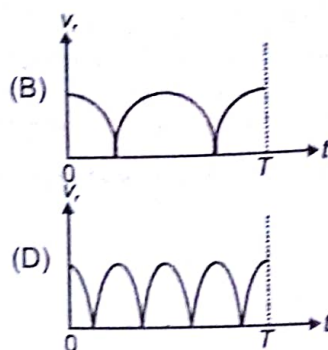
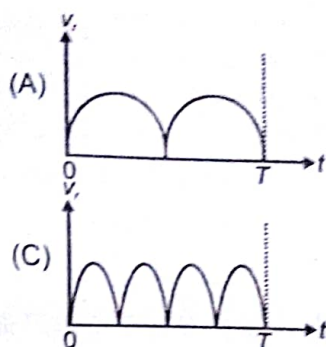
Only One Option Correct Type Questions

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[IIT-JEE-2012 (Paper-2)]

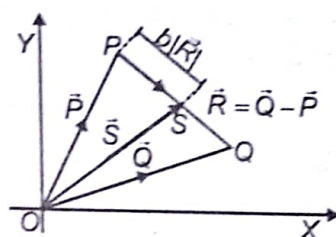


[IIT-JEE-2012 (Paper-2)]



- 3*. Three vectors $\vec{P}$, $\vec{Q}$ and $\vec{R}$ are shown in the figure. Let S be any point on the vector $\vec{R}$. The distance between the points P and S is $b|\vec{R}|$. The general relation among vectors $\vec{P}$, $\vec{Q}$ and $\vec{S}$ is

[JEE (Adv)-2017 (Paper-2)]



- (A) $\vec{S} = (1 - b^2)\vec{P} + b\vec{Q}$ (B) $\vec{S} = (b - 1)\vec{P} + b\vec{Q}$
 (C) $\vec{S} = (1 - b)\vec{P} + b\vec{Q}$ (D) $\vec{S} = (1 - b)\vec{P} + b^2\vec{Q}$

One or More Option(s) Correct Type Questions

- 4*. The coordinates of a particle moving in a plane are given by $x(t) = a\cos(pt)$ and $y(t) = b\sin(pt)$ where a , b ($< a$) and p are positive constants of appropriate dimensions. Then

[1999S]

- (A) The path of the particle is an ellipse
 (B) The velocity and acceleration of the particle are normal to each other at $t = \pi/(2p)$
 (C) The acceleration of the particle is always directed towards origin
 (D) The distance travelled by the particle in time interval $t = 0$ to $t = \pi/(2p)$ is a



5. Starting at time $t = 0$ from the origin with speed 1 ms^{-1} , a particle follows a two-dimensional trajectory in the x - y plane so that its coordinates are related by the equation $y = \frac{x^2}{2}$. The x and y components of its acceleration are denoted by a_x and a_y , respectively. Then

[JEE (Adv)-2020 (Paper-2)]

- (A) $a_x = 1 \text{ ms}^{-2}$ implies that when the particle is at the origin, $a_y = 1 \text{ ms}^{-2}$
 (B) $a_x = 0$ implies $a_y = 1 \text{ ms}^{-2}$ at all times
 (C) At $t = 0$, the particle's velocity points in the x -direction
 (D) $a_x = 0$ implies that at $t = 1 \text{ s}$, the angle between the particle's velocity and the x axis is 45°



Integer / Numerical Value Type Questions

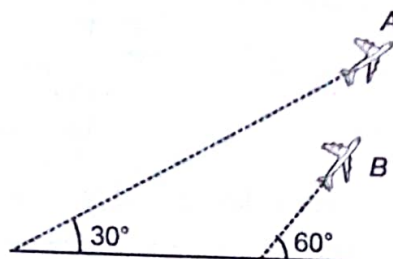
- 6*. A train is moving along a straight line with a constant acceleration a . A boy standing in the train throws a ball forward with a speed of 10 m/s , at an angle of 60° to the horizontal. The boy has to move forward by 1.15 m inside the train to catch the ball back at the initial height. The acceleration of the train, in m/s^2 , is

[IIT-JEE-2011 (Paper-2)]



7. Airplanes A and B are flying with constant velocity in the same vertical plane at angles 30° and 60° with respect to the horizontal respectively as shown in figure. The speed of A is $100\sqrt{3}$ m/s. At time $t = 0$ s, an observer in A finds B at a distance of 500 m. The observer sees B moving with a constant velocity perpendicular to the line of motion of A. If at $t = t_0$, A just escapes being hit by B, t_0 in seconds is

[JEE (Adv)-2014 (Paper-1)]



8. Two vectors $\vec{A}$ and $\vec{B}$ are defined as $\vec{A} = a\hat{i}$ and $\vec{B} = a(\cos \omega t \hat{i} + \sin \omega t \hat{j})$, where a is a constant and $\omega = \frac{\pi}{6}$ rad s $^{-1}$. If $|\vec{A} + \vec{B}| = \sqrt{3}|\vec{A} - \vec{B}|$ at time $t = \tau$ for the first time, the value of τ , in seconds, is _____.

[JEE (Adv)-2018 (Paper-1)]

9. A person of height 1.6 m is walking away from a lamp post of height 4 m along a straight path on the flat ground. The lamp post and the person are always perpendicular to the ground. If the speed of the person is 60 cm s $^{-1}$, the speed of the tip of the person's shadow on the ground with respect to the person is _____ cm s $^{-1}$.

[JEE (Adv)-2023 (Paper-1)]

10. A ball is thrown from the location $(x_0, y_0) = (0, 0)$ of a horizontal playground with an initial speed v_0 at an angle θ_0 from the x -direction. The ball is to be hit by a stone, which is thrown at the same time from the location $(x_1, y_1) = (L, 0)$. The stone is thrown at an angle $(180 - \theta_1)$ from the x -direction with a suitable initial speed. For a fixed v_0 , when $(\theta_0, \theta_1) = (45^\circ, 45^\circ)$, the stone hits the ball after time T_1 , and when $(\theta_0, \theta_1) = (60^\circ, 30^\circ)$, it hits the ball after time T_2 . In such a case, $(T_1/T_2)^2$ is ____.

[JEE (Adv)-2024 (Paper-2)]

11. A projectile of mass 200 g is launched in a viscous medium at an angle 60° with the horizontal, with an initial velocity of 270 m/s. It experiences a viscous drag force $\vec{F} = -c\vec{v}$ where the drag coefficient $c = 0.1$ kg/s and $\vec{v}$ is the instantaneous velocity of the projectile. The projectile hits a vertical wall after 2 s. Taking $e = 2.7$, the horizontal distance of the wall from the point of projection (in m) is _____.

[JEE (Adv)-2025 (Paper-2)]

Question Stem for Question Nos. 12 and 13

Question Stem

A projectile is thrown from a point O on the ground at an angle 45° from the vertical and with a speed $5\sqrt{2}$ m/s. The projectile at the highest point of its trajectory splits into two equal parts. One part falls vertically down to the ground, 0.5 s after the splitting. The other part, t seconds after the splitting, falls to the ground at a distance x meters from the point O. The acceleration due to gravity $g = 10$ m/s 2 .

[JEE(Adv)-2021(Paper-1)]

12. The value of t is _____.

13. The value of x is _____.

14. A projectile is fired from horizontal ground with speed v and projection angle θ . When the acceleration due to gravity is g , the range of the projectile is d . If at the highest point in its trajectory, the projectile enters a different region where the effective acceleration due

to gravity is $g' = \frac{g}{0.81}$, then the new range is

$d' = nd$. The value of n is _____.

[JEE (Adv)-2022 (Paper-1)]

